Consulting & Analytics Club
IIT Guwahati

StockGro

Capstone Project 2026

Time Series Analysis 2026

Important Links

Capstone Submission Link: Click Here

Virtual Trading on StockGro (Details inside)

1. Capstone Registration FORM: Click Here
2. Trading Platform link: Click Here

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Capstone Problem Statement

Data-Driven Stock Analysis using Time Series Models on StockGro

Objective

In this capstone project, you will apply time series forecasting and analysis techniques to select stocks and allocate capital within a virtual ₹10,00,000 portfolio on the StockGro platform. You will build predictive models (ARIMA, Prophet, LSTM, etc.) on historical NSE stock data, use those models to make informed investment decisions, execute those decisions on the StockGro virtual trading platform, and finally compare your model's predictions against the actual market outcomes observed on StockGro. The entire workflow — from data analysis to live virtual trading to final reporting — constitutes your submission.

Platform Background

StockGro is a virtual stock market simulator that uses live NSE/BSE data. It allows users to build portfolios, allocate funds, execute trades, and monitor real-time performance — all without any real financial risk. You will use StockGro as the execution environment to test whether your data-driven forecasts translate into actual portfolio gains.

Dataset Specification

- Source: Yahoo Finance via the *yfinance* Python library
- Period: 1st January 2021 to 31st December 2025
- Interval: Daily (1d)
- Train/Test Split: Use a minimum of the last 6 months of this period (i.e., July 2025 – December 2025) as your test/validation set. Everything before that is your training data.

Example code to fetch data:

```
import yfinance as yf
data = yf.download("RELIANCE.NS", start="2021-01-01", end="2025-12-31", interval="1d")
```

Step-by-Step Project Tasks

Follow the tasks below in order. Each task builds on the previous one.

Task 1: Stock Universe Selection

What you must do:

- Choose stocks from diverse sectors (e.g., Banking, IT, Pharma, FMCG, Auto) to ensure portfolio diversification.
- Justify each stock selection using at least one of the following data-driven approaches:
  - Rolling standard deviation to identify stocks with interesting volatility profiles
  - Seasonal decomposition to detect stocks with strong trends
  - Sector-based rationale (e.g., selecting from a high-momentum sector)

Deliverable: A written justification for your chosen stock universe.

Task 2: Data Preprocessing

Before modeling, prepare your data so it is clean and suitable for time series analysis.

What you must do:

- Handle missing values using forward fill, backward fill, or removal
- Test each stock's price series for stationarity using the Augmented Dickey-Fuller (ADF) test; apply differencing if the series is non-stationary
- Normalize or scale data if you plan to use ML-based models like LSTM
- Split the data into training (before July 2025) and testing (July 2025 – December 2025) sets

Deliverable: Clean, preprocessed time series datasets for each selected stock, ready for modeling.

Task 3: Time Series Forecasting

Build forecasting models to predict stock prices for the next 2 trading days beyond your dataset (i.e., into early January 2026, or equivalently, the 2 days of the StockGro trading window).

What you must do:

- Implement at least one (preferably two or more) of the following models:
  - ARIMA / SARIMA
  - Exponential Smoothing (SES / Holt-Winters)
  - Facebook Prophet
  - LSTM / GRU
  - CNN-based time series model
  - Any transformer-based or pretrained model
- For each model:
  - Train on the training set
  - Validate on the test set (last 6 months)
  - Forecast prices for the next 5 trading days
  - Evaluate using MAPE, RMSE, and/or directional accuracy (did the model correctly predict whether the price goes up or down?)
- Tune ARIMA using AIC/BIC criteria and validate residuals
- Optionally, combine multiple model predictions into an ensemble for better accuracy

Deliverable: Forecasted prices for each stock over the next 2 trading days, along with model evaluation metrics.

Task 4: Volatility and Trend Analysis

Beyond just price forecasts, you must understand each stock's risk profile and trend behavior.

What you must do:

Volatility Estimation:

- Compute log returns for each stock
- Analyze fluctuations in log returns to estimate short-term risk
- Optionally use GARCH(1,1) or rolling standard deviation to model time-varying volatility and forecast near-term uncertainty

Trend and Pattern Analysis:

- Apply seasonal decomposition (e.g., STL decomposition) to break each stock's price series into trend, seasonality, and residual/noise components
- Evaluate the strength and direction of the trend to understand whether a stock is in an upward, downward, or sideways phase

Deliverable: Volatility estimates and trend assessments for each stock in your universe, which will directly feed into your portfolio allocation strategy in the next task.

Task 5: Portfolio Construction and Capital Allocation

Using the outputs of Tasks 3 and 4, decide how to allocate your ₹10,00,000 virtual capital across your selected stocks. You must use at least two of the following strategies:

Strategy A — Forecast-Guided Allocation:

- Rank stocks by their predicted 5-day return from your forecasting models
- Allocate more capital to stocks with higher predicted gains and higher forecast confidence
- Optionally adjust weights based on model uncertainty or prediction intervals

Strategy B — Volatility-Aware Sizing:

- Use predicted volatility (from GARCH or rolling std dev) to assign weights inversely proportional to volatility: $w_i = (1/\hat{\sigma}_i) / \Sigma_j(1/\hat{\sigma}_j)$
- Stocks expected to be more volatile receive a smaller allocation; more stable stocks receive a larger share
- Normalize all weights to sum to 1

Strategy C — Correlation-Based Diversification:

- Compute a correlation matrix across your selected stocks using recent return windows
- Prefer combinations of stocks that are lowly correlated with each other to reduce portfolio-level risk
- Avoid concentrating capital in highly correlated stocks

Strategy D — Sector Momentum Rotation:

- Group your selected stocks by sector
- Assess which sectors are showing consistent upward momentum
- Allocate more capital to high-momentum sectors, and within each sector, use one of the above strategies to finalize individual stock weights

Deliverable: A final portfolio allocation table showing: stock name, allocated amount (in ₹), weight (%), and the reasoning/strategy behind the allocation.

Task 6: Model Comparison

Compare all the forecasting models you built in Task 3.

What you must do:

- Tabulate MAPE, RMSE, and directional accuracy for each model across all stocks
- Comment on each model's strengths and weaknesses (e.g., ARIMA is interpretable but struggles with non-linearity; LSTM captures complex patterns but requires more data and tuning)
- Discuss whether an ensemble approach improved results
- State clearly which model(s) you ultimately relied on for your portfolio decisions and why

Deliverable: A comparative model evaluation table and a written discussion.

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Task 7: Virtual Trading on StockGro

This is where you execute your data-driven strategy in the real (virtual) market.

Step-by-step instructions:

1. Register for capstone (*Registration Link*) using the provided form to gain access to the Live Event.
2. Log in at : https://stockgro.onelink.me/vNON/21jikjek
   - Log in with your Registered mobile Number
   - Go to Trackers in the app ➛ find Event : "Portfolio - Time Series Analysis 2026"
   - Make sure your StockGro app is updated, otherwise the Event Portfolio will not be visible
3. Open the Event Portfolio and start building your portfolio
4. Deploy your ₹10,00,000 by purchasing the stocks in the exact proportions determined in Task 5
5. The trading window is open Monday to Friday, **11 May 2025 – 15 May 2025 (market hours: 9:35 AM – 3:25 PM each day).** Pick any two consecutive trading days within this window as your personal execution and observation window. Execute all your trades before the market close of your chosen Day 1, and record your portfolio performance at the market close of your chosen Day 2.

Valid day pairs you can choose from:

- **Day 1: Mon 11 May → Day 2: Tue 12 May**
- **Day 1: Tue 12 May → Day 2: Wed 13 May**
- **Day 1: Wed 13 May → Day 2: Thu 14 May**
- **Day 1: Thu 14 May → Day 2: Fri 15 May**

To confirm enrollment: if "Portfolio – Time Series Analysis 2026" appears in your Trackers, your registration is successful.

Tutorial video: Click Here

Task 8: Performance Tracking and Comparison Against Your Predictions

This is the most critical task — comparing what your models predicted against what actually happened in the market. Throughout this task, "Day 1" and "Day 2" refer to whichever consecutive day pair you selected in Task 7.

Step 1 — Record Actual Outcomes:

- At the market close of your Day 2 (3:25 PM), note the actual closing prices of all your selected stocks from the StockGro app
- Record your portfolio's total return, individual stock returns, and any volatility observed across the two days

Step 2 — Compare Predictions vs Reality:

- For each stock, place your model's forecasted price for Day 1 and Day 2 alongside the actual prices observed on StockGro
- Compute the prediction error (MAPE or absolute % error) for this live two-day window
- Check directional accuracy: did your model correctly predict whether each stock went up or down?

Step 3 — Portfolio Performance Analysis:

- Report your total portfolio return (%) over your chosen two-day window
- Identify which stocks contributed positively and which dragged performance
- Assess whether your chosen allocation strategy from Task 5 held up under actual market conditions

Step 4 — Reflect:

- Did your model predictions align with actual market movement?
- Did higher-confidence stocks actually perform better?
- Was your volatility estimation accurate? Did volatile stocks behave as expected?
- What would you change in your model, preprocessing, or allocation strategy if you were to redo this?

Deliverable: A structured comparison table (predicted vs actual prices and returns) and a written reflection.

Final Submission Requirements :

Your submission must include all of the following:

1. Python Notebook(s) (.ipynb):

- Full pipeline: data fetching ➛ preprocessing ➛ modeling ➛ forecasting ➛ portfolio construction
- All code must be runnable and clearly commented
- Reference sample notebook provided: stock_price_prediction_using_yfinance[1].ipynb

2. Report (maximum 10 pages):

Your report must cover the following sections in order:

- Methodology and models used
- Stock selection rationale (Task 1)
- Data preprocessing steps (Task 2)
- Forecast results with confidence intervals (Task 3)
- Volatility and trend analysis findings (Task 4)
- Portfolio composition and allocation rationale (Task 5)
- Model comparison and accuracy evaluation (Task 6)
- StockGro execution summary — trades placed, capital deployed (Task 7)
- Predicted vs actual outcomes — this is the core of your report (Task 8)
- Reflections: what worked, what didn't, and what you would improve

3. Optional Bonus — Visual Dashboard:

Build an interactive or static dashboard that includes:

- Forecast plots: actual vs predicted prices for each stock
- Portfolio allocation chart (pie or bar chart by stock and sector)
- Correlation heatmap (if using diversification strategy)
- Trend and volatility graphs supporting your strategy rationale

Submission Link: Click Here

All The Best
~Team C&A